

DSC 424 Final Project

NBA RAPTOR Dataset

Analysis by Jack Daly and John “Jack” Leniart

Table of Contents

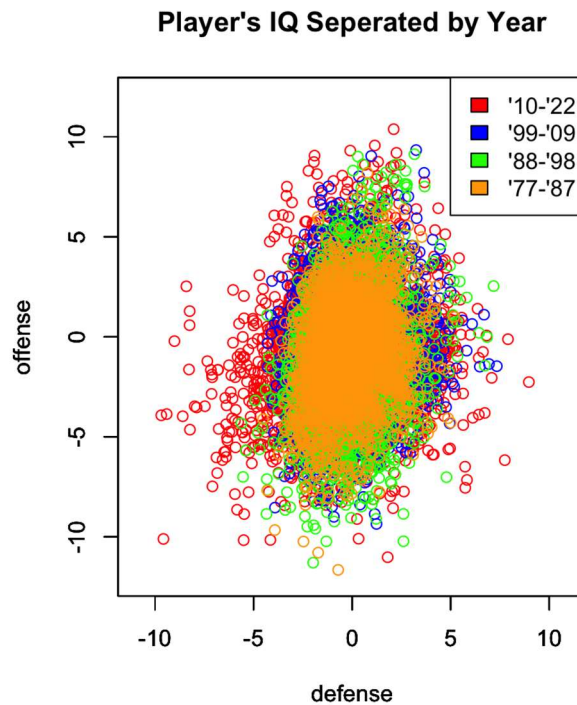
Non-Technical Summary	Page 3
Technical Summary	Page 7
I. Exploratory Analysis	Page 7
II. Regression	Page 8
III. Principal Components Analysis	Page 10
IV. Conclusion	Page 13
Appendix	Page 15

Non-Technical Summary

Our analysis of the Raptor dataset allowed us to draw some interesting conclusions revolving around the evolution of basketball at the professional level and opened a gateway that leads to a deeper understanding of sports data in general.

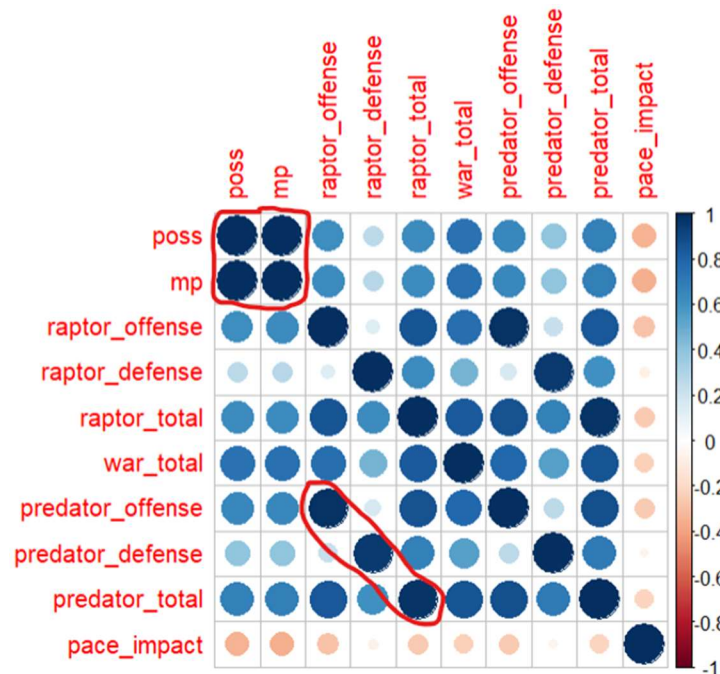
One of the most important lessons learned during this project was the idea that: “the more data you have, the easier the job is” is not always true. After exploring the Raptor dataset, which had stats for every single player that has been in the NBA since the NBA and ABA merger in 1977, it became clear that segmenting the data into chunks and looking at smaller pieces at a time would be much easier. Of course, this stems from lack of domain knowledge in working with time series, but regardless, this practice helped immensely. The segments we ended up working with were: 1977-1987, 1988-1998, 1999-2009, and 2010-2022. Four different “eras” of the NBA that all contained very unique data. This raptor dataset has so much data so the idea of model building around certain variables, and looking for deeper connections between these variables that might not be visible at first was the goal, and for that reason, the two lines of analysis we executed were: Regularized Regression and Principal Components Analysis.

Before trying any regularized regression, the original idea we had was to try to fit an model using the Ordinary Least Squares (OLS) method. This went very poorly. As one may assume, with so much data there was a lot of multicollinearity and overfitting going on. The OLS model had a R-Squared value of 1, which more often than not indicates that overfitting is occurring. A great way to combat this overfitting is to use Regularized Regression. Because a big part of the reason we wanted to use regression was to be involved in the model pruning process, we selected LASSO to use. This would allow us to see specifically what variables are correlated and involved with our selected independent variable. The dependent variable we selected for LASSO regression was `raptor_offense`, a measure of a player's offense abilities. This was selected because it was related to a lot of other variables and also let us measure the “roundness” or “IQ” of a player. We decided that the IQ of a player was determined by a mixture of a player's offensive and defensive rating, if they had both good offense and defense, they had a high “IQ”. We saw that this theory was a little broken as in later years we see that defense is completely dropped from the models, which lead us to a very interesting conclusion: A growing trend in the NBA is “specialism.” Although a player might not be great on both sides of the court, creating a roster with a mixture of offensive and defensive specialists is something that is way more common in the modern NBA than in the past.



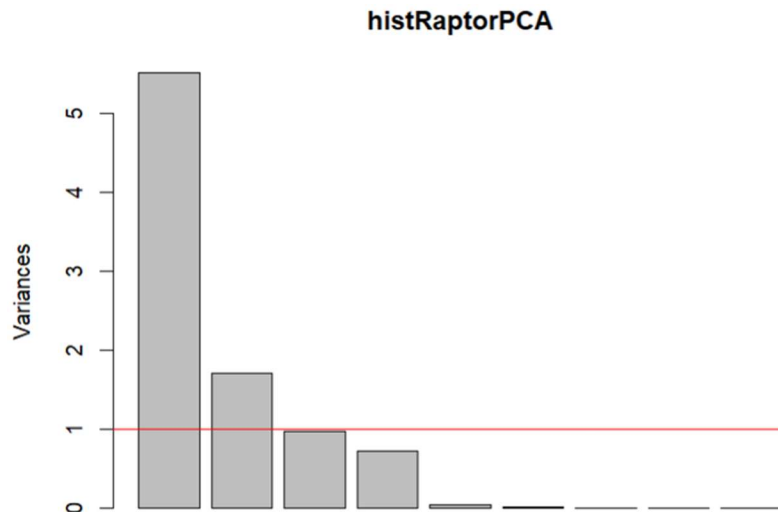
As we can see, as time goes on, players generally move farther away from the center of the graph and spread out towards the corners. We see almost no orange points ('77-'87) that far from the center, while we see plenty of red points ('10-'22) all over the place! This was discovered thanks to iterative model building and observations from the results of regression.

We also performed Principal Components Analysis (PCA) on our dataset. We thought PCA would be useful to explore because we suspected there was multicollinearity present in our dataset. PCA is a method of analysis that can be used to circumvent the problems that multicollinearity presents. Our suspicions of multicollinearity were confirmed when we checked the correlation matrix for the numeric variables in our dataset.



The correlation matrix illustrates the correlation value using colored circles. The larger and darker the color of the circle, the stronger the correlation between those two variables. As you can see in the matrix, there are multiple pairs of variables (circled in red) that are nearly perfectly correlated.

After calculating the principal components, we looked at the scree plot, which is a histogram that shows how much of the total variance in the data is captured by each of the components. This helps us determine which components are significant and which are just noise.



At first, we decided to focus on the first three components because the fourth component falls short of the variance threshold (indicated by the red line in the graph) that is commonly used. However, using a technique called parallel analysis, we determined that it would be appropriate to include the fourth component as well.

When analyzing the linear combinations of the variables that made up each component, we concluded that the first four components can be used to describe a player's overall value, their balance of defensive and offensive skill, and how they affect the pace of the game.

Technical Summary

I. Exploratory Analysis

Our work on this project started with some exploratory analysis of the dataset in the “historical_RAPTOR_by_player.csv” file. We wanted to gain an understanding of the data included in this dataset and take care of any clean-up that we felt was needed.

The original file contains 15 different variables and 19,159 observations. However, only 12 of the 15 variables are numeric. For our analysis, we wanted to focus on those numeric variables.

There was also a bit of redundancy among the variables in the dataset. The variables wins above replacement for the regular season (war_reg_season) and wins above replacement for the playoffs (war_playoffs) are both included in the calculation of another variable (war_total). So, we decided to drop both war_reg_season and war_playoffs from the dataset.

The dataset contains observations from every NBA season from 1977 to 2022. Those who are fans of the NBA know that the game has changed drastically over the years. Comparing individual players or the style of play from two different eras is like night and day in some respects. Because of this, we divided the dataset into four different eras: 1977-1987, 1988-1998, 1999-2009, and 2010-2022.

When looking at the summary of all the numeric variables, we noticed that the values for the variables possessions played (poss) and minutes played (mp) were much larger than the range of values for the other 10 numeric variables. It was important to note this discrepancy because it is a strong indicator that we should scale the data before performing certain types of analysis.

Another thing we learned about the dataset was that it contained observations for every single player who appeared in an NBA game during the timeframe (1977-2022). This means the dataset includes observations for players who played only one possession or only one minute during the course of an entire NBA season.

We felt that including the observations with a small sample size of playing time would not help our analysis. On the contrary, we thought that those observations could hinder our analysis. The small sample size of playing time could have skewed the results of the other metrics in the observation.

We decided to drop any observations where the minutes played (mp) was less than 99. This number was chosen a bit arbitrarily. We felt it was a good threshold because some of the metrics in the dataset are calculated per 100 possessions and all of the players above the threshold had greater than 100 possessions played (poss). In total, we dropped 1,896 observations from the dataset.

After dropping those observations, the range of the values for all of the raptor and predator variables (offense, defense, and total) got smaller. This tells us that the observations that had a smaller sample of playing time yielded some extreme values on both ends of the spectrum for those variables.

II. Regression

Although regression is one of the most basic forms of data analysis, it can also be called one of the most essential. Regression allows us to take specific pieces of our data and see their relationships (or lack thereof) with the rest of the data. Regression also interested us as an interesting direction to elevate this project to the next level would be to make a model or simulator that could predict: Player ratings, outcome of individual games, or entire seasons. Regression is the stepping stone to this type of simulation, and therefore was a great choice to dive into.

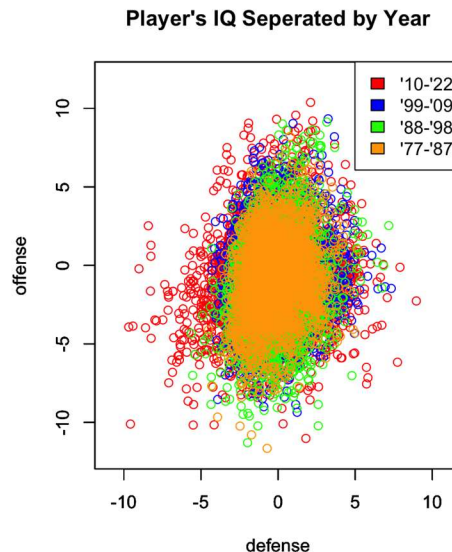
To start looking into regression, a general Ordinary Least Squares fit was made with just a full fit of the data. Of course, as most fullfits go, this model was very messy. After testing for multicollinearity and looking at VIF values, it was clear that it was present, so the idea of regularized regression came to mind, but before moving forward we still ran through all of the steps of cutting out variables. It seemed like regardless of what was done to the OLS model, the R squared value remained at a solid 1. This is clear evidence of overfitting, and we quickly moved on to Regularized regression. Because we chose regression to get a better idea of how the variables relate to one another in the first place, we chose LASSO as the model building process is still very intimate. During our exploration of OLS, we went through tons and tons of models to try to make one that would make sense with no avail, but with the use of LASSO we were able to generate some models that actually provided us with some good weight. The models we got were also made with a prediction using a test and training set. It was important to use test and training sets so we would ensure that the results we are seeing have some sort of validation to them, which we did observe. The training set was made with 2/3s of the data and the rest of the data was taken for the test set, which then produced our final models after running LASSO with a predictor. They are shown below:

Independent Variable: Raptor Offense	
Model 1 Poss: -0.00000311 Raptor Defense: -0.761 Raptor Total: 0.750 Predator Offense: 0.246 Predator Defense: -.00453 Pace Impact: -0.0224	Model 2 Poss: -0.00000323 Raptor Defense: -.714 Raptor Total: .697 Predator Offense: .301 Pace Impact: -.0243
Model 3 Poss: -0.00000264 Raptor Defense: -.735 Raptor Total: .739 Predator Offense: .255e Predator Defense: -.00558 Pace Impact: -.0241	Model 4 Raptor Defense: -.830 Raptor Total: .835 Predator Offense: .158 Pace Impact: -.009
Average mse: 4.36	

As you can see from the top of the graphic above, the independent variable was Raptor Offense, and all of these models had an average MSE of 4.36 so we are getting a fairly accurate view of each decade and compare and contract them as they were all run through the exact same steps for LASSO. The first few models gave us some great betas but as we move on this begins to change. Model 1 represents the model for '77-'87, model 2 represents '88-'98, model 3 represents '99-'09 and model 4 represents '10-'22. In model 1 we can see that there is a fairly strong correlation between raptor offense and both of the defensive and offensive ratings. Model 2 we can see that predator defense was dropped but otherwise is fairly similar. Model 3 brings predator defense back, but model 4 shows the most interesting results. The first thing to note is that the possessions variable is included in every single model and has some of the lowest betas showing that it is very important to the model, but model 4 drops this. It makes sense that a player's offensive rating would stem closely with their possessions, the more possessions they have the more chances they have to make plays. Another interesting point to look at within model 4 is that raptor defense is hardly in the model with a very high beta coefficient. So why would possessions fall out of the model? And why would raptor defense have such a high beta? This led to the suspicion that specialization is occurring.

The NBA is going through a large phase of something called "positionless basketball." Although plenty have critiqued it, it has shown to be a great way to really utilize every single player on a roster, and get around their shortcomings as a player, and essentially uses "specialization" Specialization in the NBA is the act of using players as specialists rather than an all around player. The example I like to look back to is Chicago's own DeMar Derozan. Derozan does not exhibit any great defensive efforts but is generally not put in many positions in which he has to. Rather than guarding the player in the same position as him on the opposing team, demar is assigned to who he can reasonably guard, and allows a better defensive player to guard someone who is a great offensive player. Effectively removing Demar's "Position." With this idea, it makes more and more sense for NBA teams to scout players who are not amazing in all aspects of the game. This is exactly why in the modern NBA, having a great defensive rating does not mean you have a great offensive rating. As for the possession dropping out, some of the best offensive specialists could be someone like Seth Curry (Steph's brother) who for a

while was a bench player, not getting many possessions, but still providing a great deal of offense to his team and therefore having a great offensive rating with low possessions. This idea of specialization can clearly be seen in the plot below which was generated as a result of the trying to rationalize the results of the regression:



All in all, with as big as a trend positionless basketball is, I would say that the regression that was run on this dataset was a success in showing this trend and therefore proving that basketball, at a professional level, is evolving right before us. Even more so, going through large changes **right now** more so than ever.

III. Principal Components Analysis

As we discussed in the non-technical summary, we chose Principal Components Analysis (PCA) as one of our lines of analysis because we suspected there was multicollinearity present in the dataset. Our initial analysis of the data confirmed that suspicion. Before running PCA, I removed the `war_total` variable from the dataset I was working with as I figured it was one of the best candidates to serve as the dependent variable in this analysis.

Based on the ranges of values for the variables in our dataset, I believe it would be best to run scaled PCA. However, I wanted to check unscaled PCA first to test this assumption. As expected, running unscaled PCA produced only one significant component, which accounted for 99.97% of the variance present in the data. As I discussed earlier, this result is due to the fact that two of our variables (`poss` and `mp`) have much larger values than the other 10 numeric variables.

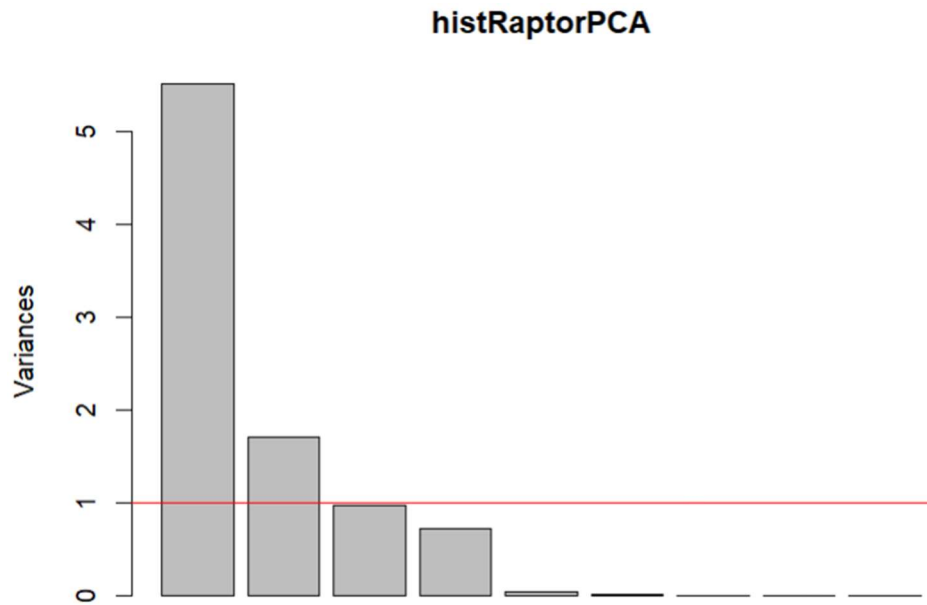
If we run scaled PCA, we get much more useful results. You can see from the output below that the variance captured by the components is much more spread out.

```
> summary(histRaptorPCA)
Importance of components:

```

	PC1	PC2	PC3	PC4
Standard deviation	2.3485	1.3092	0.9903	0.84872
Proportion of Variance	0.6129	0.1904	0.1090	0.08004
Cumulative Proportion	0.6129	0.8033	0.9123	0.99229

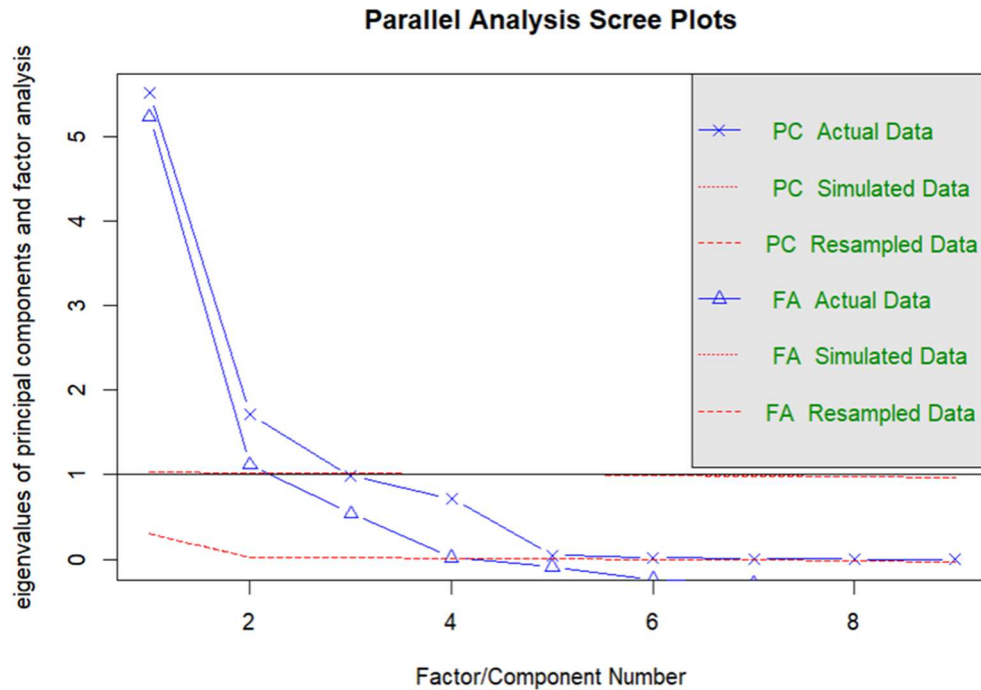
Next, we needed to choose how many of the components we would focus on. To do so, we looked at the scree plot. There are a few different methods we used to help determine the number of components we were going to focus on.



First, we looked for a “knee” in the scree plot, which is spotted by looking for a leveling off in variance. There does appear to be a knee after the fourth component. Next, since we are using scaled data, we looked for components that fell short of the variance = 1 threshold. The third component is just shy of that threshold and the fourth component falls well short of it.

At this point, the decision about the number of components we should choose was still unclear. So we ran Horn’s Parallel Analysis to help us make that decision.

In parallel analysis, we rerun PCA using a randomized dataset with the same dimensions as our dataset. We do this to see how many principal components would be selected from truly random data. The result of the analysis was that we should choose the first four principal components.



After choosing the number of principal components to select, the next step was to look at the make-up of each component and analyze their meaning. Each component can be represented by a linear combination of the variables in the dataset. We can look at the coefficients for each variable to gain an understanding of what each component is measuring.

Rotation (n x k) = (9 x 9):

	PC1	PC2	PC3	PC4
poss	0.3465284	0.18602117	0.3252985	-0.48792721
mp	0.3484870	0.18334857	0.3278724	-0.47733495
raptor_offense	0.3609779	0.29709170	-0.3188212	0.18623314
raptor_defense	0.2373691	-0.61631964	0.1212062	0.09626577
raptor_total	0.4077545	-0.08772155	-0.1875861	0.19670391
predator_offense	0.3688215	0.27737364	-0.3198459	0.12579796
predator_defense	0.2773540	-0.56421189	0.1002660	-0.03284929
predator_total	0.4135041	-0.08115191	-0.1856728	0.07636851
pace_impact	-0.1470441	-0.23262448	-0.6988026	-0.65475165

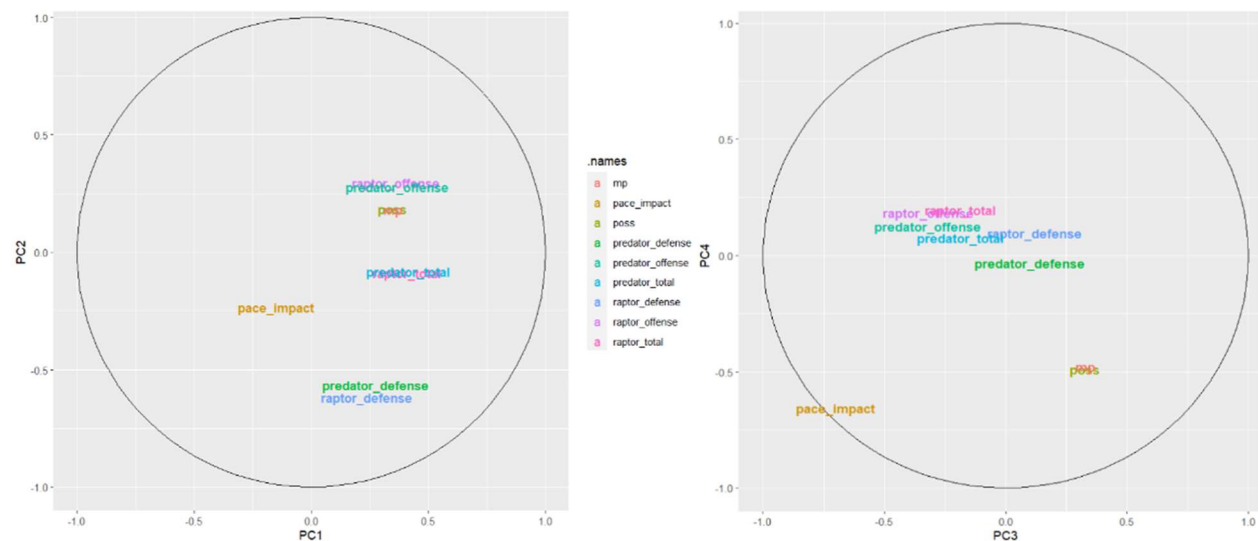
The first component (PC1) tells us about a player's overall value based on the metrics in our dataset. The largest contributions to this component come from both of the total variables. It also has notable contributions from both of the offense variables, possessions played, and minutes played.

The second component (PC2) tells us if someone is more of a defensive player or more of an offensive player. The largest contributions to this component come from the two defense variables, which both have negative coefficients. This component captures a negatively correlated relationship between the defense and offense variables.

The third component (PC3) tells us how a player affects the pace of the game specifically from an offensive perspective. The largest contribution to this component comes from the pace impact variable. It captures a negatively correlated relationship between pace impact and both possessions played and minutes played. It also captures a positively correlated relationship between pace impact and the offensive variables.

The fourth component (PC4) is similar to PC3 in that it also tells us how a player affects the pace of the game. Its largest contribution comes from the pace impact variable, and it captures a positively correlated relationship between pace impact and both possessions played and minutes played.

Another tool that can help us understand the make-up of these components is PCA Plot. This function allows us to generate a plot of the variable coefficients for two adjacent components.



In the PCA Plots above, you will notice that there are pairs of variables that are close together in both plots. These are the same pairs of variables we identified when looking at the correlation matrix for this dataset. Unsurprisingly, the highly correlated variables have very similar coefficients in each of the four components.

IV. Conclusion

In conclusion, the analysis we were able to do on the Raptor dataset was very successful in the sense that we were able to capture a clear evolution of the game on an individual player ratings level, along with breaking down how these individual player ratings come to be. Rating players is nothing new and a dataset that includes player ratings for individual stats has a lot of hidden data in it, which we were able to extrapolate with Principal Component Analysis. Regression proved to be a great line of analysis for this dataset as well as it showed us that over time, it is clear to see that the game of basketball changed quite a bit, emulating the results of a time series analysis. We saw a clear drop off in the correlation between offense and defense, a

relationship that in the modern NBA seems foreign, but clearly was an important relationship within the history of the NBA.

Perhaps more importantly that the conclusions which were made from our findings, are the lessons learned from performing data analysis from scratch. These lines of analysis were by no means executed perfectly on the first try, and they took a lot of time and effort. One of the original lines of analysis we had decided on was cluster analysis. The theory behind cluster analysis was to see if we could separate the data in terms of positions, which originally prompted discussion of “positionless” basketball, but ultimately failed because of a lack of understanding of the data. Finding a player's position based on their overall ratings does not make very much sense (especially in the modern NBA) as just because someone plays center, does not make them a better defense or offensive player, and we found this clear by trying for hours on end to make this specific cluster analysis work. This is an example of bias that was involved in our research as well, and that is exactly why we did not include any of that research in the actual findings. Trying to make something work rather than just probing different areas of the data and taking an unbiased look at the data is bad practice. Regardless of how long it took, that was one of the best lessons to learn first-hand.

Another important lesson learned was the intricacies of performing data analysis with a team. Although our group only consisted of the two of us, we still had to have a lot of synergy and good communication in order for our lines of analysis to align and work together. Constantly sending files back and forth while still trying to make it easy for one another was difficult to balance with still producing a good report. Working with a team is very common in the professional world of data science, and of course has a lot of positives that come with it (which we used to their full extent) but being aware of the difficulties that come with it is something to be aware of moving forward in this space.

All in all, the Raptor dataset was a great choice for us as having the proper domain knowledge to go along with the data that we were researching was critical and made the analysis something to look forward to. The game of basketball is changing right before our very eyes, and to capture such changes with legitimate data was beyond interesting. These types of changes are happening all around us, not just in basketball, and it is clear that with the knowledge we are obtaining, we can capture these changes very precisely.

Appendix

Individual Reports:

Jack Daly

Because we had a team of two, Jack L and I worked pretty close together on the majority of the project. We both took on one line of analysis and the one which I took on was regression. I felt like regression was a legitimate option here as model building over multiple different datasets might somewhat emulate a time series and how variables are correlated to one another over time. Jack and I agreed on this and when I ran into trouble or became confused on something we usually talked it over together. After I had some good analysis, I made all of the regression plots and generated a nice graphic to hold all of my models. Jack L set up the slideshow as a big outline of what needed to be done and then I went through and filled in my parts, and we split up the parts that we both worked on evenly. I thought it was important to get a few examples of players to cite during the paper and the presentation so I found a few that fit the mold we needed and discussed them when appropriate.

I think both of our lines of analysis contributed a lot to the presentation. The regression allowed us to really break down the data into eras, which for this data set we found to be very important. We really wanted to find the evolution of the game and thought that breaking down the dataset would allow us to do so, and regression was the tool we used to do so. Regression over time truly emulated a time series and we could see how the models for the exact same dataset taken over different decades changed so much. Possessions dropping out was a big one specifically as it was such a leading coefficient in the other models and all the sudden just dropped out. This told me that something was up. Again, I truly believe that if we did not have the domain knowledge that we do in basketball and in the modern NBA, it would be pretty hard to draw these conclusions. I asked myself “why do possessions not matter” and was able to think of some examples of players who do not get a lot of possessions but still contribute so much to the team in a plethora of different ways.

I also think regression helped us rationalize PCA a bit. Although we did not run regression on top of the PCA data, there was obviously a large connection in finding how variables play a role on one another and then using regression to find out why that may be.

John “Jack” Leniart

Jack D and I were in constant communication throughout the course of this project to ensure that we made the necessary progress to meet each of the milestones. As NBA fans, we were both very interested by the dataset and excited to get started.

Outside of our two main lines of analysis, I thought the rest of the work we completed was divided evenly. We each worked on initial exploratory analysis separately and then discussed our findings together. I started my analysis on the dataset in the “latest” file, and Jack D began

his analysis using the dataset in the “historical” file. After discussion, we both thought that it could potentially be more interesting to focus our analysis on the “historical” dataset. So that is what we did moving forward.

During my exploratory analysis, I noticed our dataset contained observations with a low value for possessions played and minutes played. I thought it would be best to drop those observations from our dataset as the small sample size of playing time could have skewed the other metrics in those observations. I shared this idea with Jack D and he also agreed that it would be beneficial to remove those observations.

We also discussed wins above replacement variables in the dataset. We thought the `war_reg_season` and `war_playoff` variables were redundant since both variables were used in the calculation of `war_total`.

My proposed line of analysis was Principal Components Analysis. While completing my exploratory analysis, I noticed the dataset had variables that sounded like they would be strongly correlated. Checking the correlation matrix showed me that there was some multicollinearity among the variables. Earlier in this course, we learned that PCA is a method that can be used to deal with multicollinearity. It seemed like a perfect fit for this dataset.

I ran scaled PCA using the `prcomp()` function. To decide on the number of components to select, I looked for a knee in the scree plot, checked which components fell below the variance = 1 threshold, and ran Horn’s Parallel Analysis. Ultimately, the results of those three suggested we should select the first four components.

To interpret the first four components, I looked at the coefficients for each of the variables and also used the PCA Plot functions to help visualize the contributions from each variable on the components.

Jack D and I were able to meet in-person for a working session before we submitted milestone 5 for this project. We had a very productive meeting where we talked through each of our proposed lines of analysis. That meeting helped us create a blueprint for the remainder of the project.

For our presentation materials, I created a shared Google Slideshow so that we could both add information and edit on our own time. We each added materials for our lines of analysis (regression for Jack D and PCA for myself), and then we divided the remaining content (e.g. introduction, conclusion, exploratory analysis, proposed future analysis). We followed a similar pattern when composing this final report. We each provided all of the information for our line of analysis and we agreed to divide the remaining section of the report.

This project helped me become much more comfortable with PCA, a topic I struggled with a bit when I first learned about it. I was quite proud of myself when, during exploratory analysis, I had an “a-ha” moment realizing that PCA would be a useful line of analysis for this dataset. Being

able to practice this method of analysis with real data was very engaging and helped solidify everything I had learned.

Another takeaway from this project was that real data analysis is not always a straight line from point a to point b. Jack D and I had to pivot a few times during the course of this project as we hit roadblocks or thought of ways improve our analysis.